

SNP — Site Visit Working Agenda

Personal prep document. Not for distribution.

Date	_____	Location	_____
SNP attendees	_____	Ours	_____

BEFORE YOU GO

Do not signal the second-source work. Every technical question here has a truthful cover: we are formalising internal spec documentation. That is genuinely true and it explains all of it.

Strategic frame worth testing today: our research concluded SNP is a PVA-cooking specialist — it is their product line, not a side job — which is exactly why \$0.75/lb has been hard to beat (market base ~\$1.40/lb). An outside second source costs +\$11k–\$43k/yr. So the real question is whether redundancy INSIDE SNP — second line, second site, staged resin, safety stock — buys more continuity per dollar. If it does, it reframes the whole initiative.

One discipline: you will want to fill silences by explaining why you are asking. Don't.

OUR NUMBERS — quick reference

Volume: ~1,300 lb/week · 450 lb/drum · ~150 drums/yr

Price: \$0.75/lb delivered ≈ \$337.50/drum ≈ \$50,700/yr

Spec: 10–12% solids · <2,500 cps centipoise (#3, 10 rpm, 25 °C)

Shelf life: 18 days · fresh material typically <1,000 cps

Open gaps we can close today: pH target · viscosity lower bound · colour standard · exact resin grade · cleanup procedure · hazard classification · drum spec · test method

PART A — YOUR TEAM'S ASKS

1 · Drum recycling

- Do you run a returnable / reconditioned drum program? Could empties go back on the SAME truck that delivers (backhaul, no added freight)?
- Who owns cleaning, and does our residue qualify as RCRA-empty? (RCRA = Resource Conservation and Recovery Act, the US rule that defines when a container counts as 'empty'.)
- Does returning drums reduce our per-drum price — and by how much?
- Do you offer recycled-content or reconditioned drums as an option?
- What drum do you use for us today — poly or steel, open or tight head?

Our angle: water-soluble PVA rinses with hot water, unlike solvent or hazardous residues — that makes our drums unusually good reconditioning candidates. ~150 drums/yr also maps to Intuitive sustainability targets, which is an internal win to report.

2 · CoA vs CoC — Tom's item ★ highest-value item of the visit

- Move us from Certificate of CONFORMANCE (CoC — asserts it passed) to Certificate of ANALYSIS (CoA — gives the measured numbers). Fields: lot #, production date, % total solids (%TS), final pH, viscosity.
- ★ Your exact viscosity METHOD: instrument series (LV / RV / HA / HB — the four Brookfield torque-spring ranges, from Low to High Viscosity), spindle designation, RPM (revolutions per minute), temperature + tolerance, and reading time.
- Can we get 12–24 months of historical CoA (Certificate of Analysis) data?
- Do you keep retained samples, and for how long?

Why the method matters: 'spindle #3, 10 rpm, 25 °C' is NOT a complete method. LV-3 and RV-3 are different geometries and impose different shear rates, so they give different numbers on a shear-thinning fluid like ours. SNP's method is the de facto reference standard — and right now we don't know it.

Why the history matters: one dataset closes three gaps at once — the missing pH target and tolerance, a real viscosity LOWER bound, and normal batch-to-batch variation.

3 · Other hydrogels — the engineers' interest

- What other water-soluble polymer solutions do you make? PVP (polyvinylpyrrolidone), PEG (polyethylene glycol), cellulose — HEC (hydroxyethyl cellulose) / HPMC (hydroxypropyl methylcellulose), alginate, polyacrylamide, gelatin, carbomer.
- Can you formulate blends — PVA (polyvinyl alcohol) / PVP, PVA/gelatin? Do you have R&D (research and development) / formulation support to help develop variants?
- To tune mechanical properties, what levers do you see: PVA molecular weight (MW) and hydrolysis grade, solids loading, blends, or chemical crosslinking (borate, glutaraldehyde) vs our physical freeze/thaw route?
- Do you have freeze/thaw cryogel experience? Cycle count, freeze rate and solids all tune stiffness — they may know things we don't.

Strategic read: if SNP can make several hydrogel chemistries they stop being a commodity supplier and become a development partner. That is an argument for deepening the relationship rather than diversifying away from it.

4 · Their volume, capacity, and where we sit

- Total plant capacity — how many reactors, what sizes, one site or several?
- What fraction of your output are we? (We're ~150 drums, ~34 tons/yr — likely a very small account.)
- Where do we sit in your customer mix? Does our weekly cadence cause friction, or fit neatly?
- Lead time normally vs at peak. Is there a busy season?
- ★ If your primary reactor goes down, what happens to us?

Knowing our share honestly tells you two things at once: how much leverage you have, and how easy you are for them to serve. The last question is the redundancy question, asked innocently.

PART B — MY ADDITIONS

5 · Business continuity (the strategic one)

- Do you have a documented BCP (business continuity plan)? A second line or second site that could run our product?
- Could you pre-stage or hold safety stock of our resin, so an upstream disruption doesn't stop our drums?
- Would you support us holding a buffer — and what actually limits that?

6 · Shelf life — why 18 days?

- ★ What actually limits the 18 days: microbial growth, viscosity build, phase separation, or pigment settling?
- What is the viscosity at time of MANUFACTURE vs what we measure on receipt? Confirm the build behaviour.
- If we restructured the spec as fresh target + end-of-life ceiling, could the usable window be longer?

This is upstream of a lot of our economics. If the limit is viscosity build rather than microbiology, a longer window would change order cadence, allow larger batches, and materially lower the cost of any second source. We've treated 18 days as a given and never asked what drives it.

7 · Close our open spec gaps — they can answer every one of these

- Exact resin grade and manufacturer. Our notes say 'Selvol S-1551F-D or equivalent' but that code doesn't match Sekisui's usual catalogue numbering. (May be treated as proprietary — frame as continuity documentation.)
- pH target and tolerance — currently missing entirely from our spec.
- Colour standard — is there an L*a*b* target (the CIE colour-space coordinates: Lightness, red–green, blue–yellow) with a Delta-E tolerance (the single-number total colour difference), or is a retain used as the visual standard?
- Cleanup procedure — theirs, in writing.
- Hazard classification: flammable / corrosive / toxic / MARINE POLLUTANT? (The biocide is aquatic-toxic; whether the diluted product trips the threshold is a real question and their SDS — Safety Data Sheet — answers it.)
- Who authors the SDS (Safety Data Sheet), and can we have the current version?

8 · Commercial — light touch

- Volume tiers — is there a break at higher annual volume?
- Would an annual agreement buy price stability? How do raw-material moves pass through?
- Payment terms. Any freight optimisation if we adjusted order size or frequency?

Frame as budgeting and forecasting, not negotiation.

9 · People

- Named technical contact + their backup. QA (quality assurance) contact for CoA questions.

Single-point-of-contact risk is real for a small account.

WHAT TO WALK OUT WITH

- ☐ Their exact viscosity test method, written down — the reference standard everything else is judged against.
- ☐ A commitment to CoA instead of CoC, with the fields agreed.
- ☐ Historical CoA data, or a firm promise of it — closes the pH and viscosity-bound gaps in one step.
- ☐ A straight answer on what actually limits shelf life.
- ☐ A read on whether internal redundancy at SNP is real — if it is, it is likely cheaper and faster than anything we've been chasing.
- ☐ Named technical + QA contacts.

FOLLOW-UPS / ACTIONS

Action	Owner	By when

ACRONYMS USED IN THIS DOCUMENT

BCP — Business Continuity Plan	cGMP — current Good Manufacturing Practice (FDA rules for drugs/some devices)
CoA — Certificate of Analysis — the measured test results for that lot	CoC — Certificate of Conformance — asserts it met spec, without the numbers
cps / cP — centipoise — the unit of viscosity (water $\approx$ 1 cps)	Delta-E — the single-number total colour difference between two samples
FDA — US Food and Drug Administration	HEC / HPMC — hydroxyethyl cellulose / hydroxypropyl methylcellulose
HMW — high molecular weight	IBC — Intermediate Bulk Container — a 275–330 gallon tote
ISO 9001 — the general quality-management standard (not industry-specific)	ISO 13485 — the medical-device quality standard — NOT required for us
L*a*b* — CIE colour-space coordinates: Lightness, red–green, blue–yellow	LV / RV / HA / HB — the four Brookfield viscometer torque-spring ranges
MOQ — Minimum Order Quantity (differs from minimum BATCH size)	MW — molecular weight
NDA — Non-Disclosure Agreement	PEG — polyethylene glycol
PVA / PVOH — polyvinyl alcohol — both abbreviations mean the same polymer	PVP — polyvinylpyrrolidone
QA — Quality Assurance	R&D — research and development
RCRA — Resource Conservation and Recovery Act — defines an 'empty' container	RPM — revolutions per minute — the viscometer spindle speed
SDS — Safety Data Sheet	%TS — percent total solids — the non-water fraction
316L / 304 — stainless steel grades; 316L resists chloride pitting (we have salt)	

NOTES

